Piano audio to bluetooth v1

Description

This is a device that is designed to connect to a digital piano with aux out and a usb-midi. It will allow user to connect piano and output bluetooth audio and midi. This device may also include a screen and controls. The planed primary components are the stm32F446, pcm1808 and cc2456.

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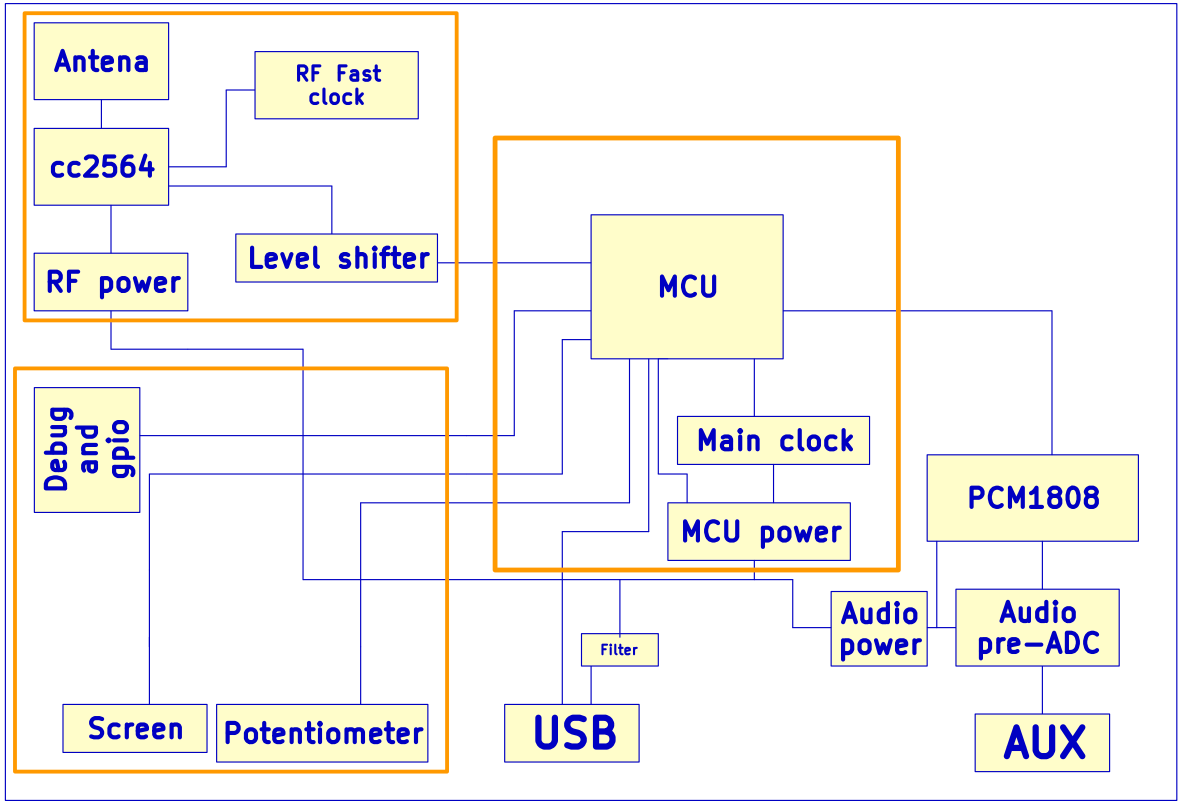

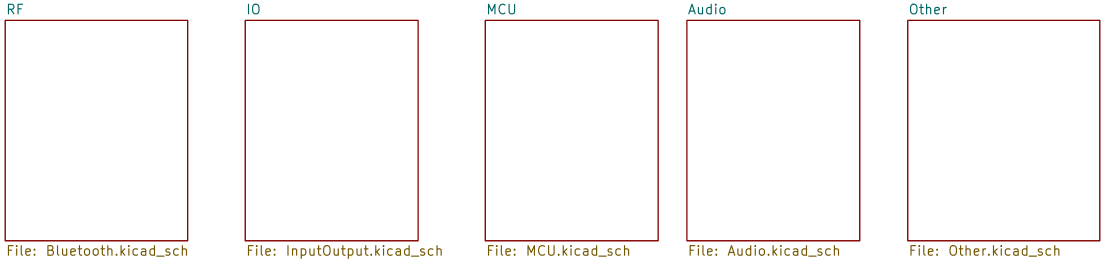

Personal Project
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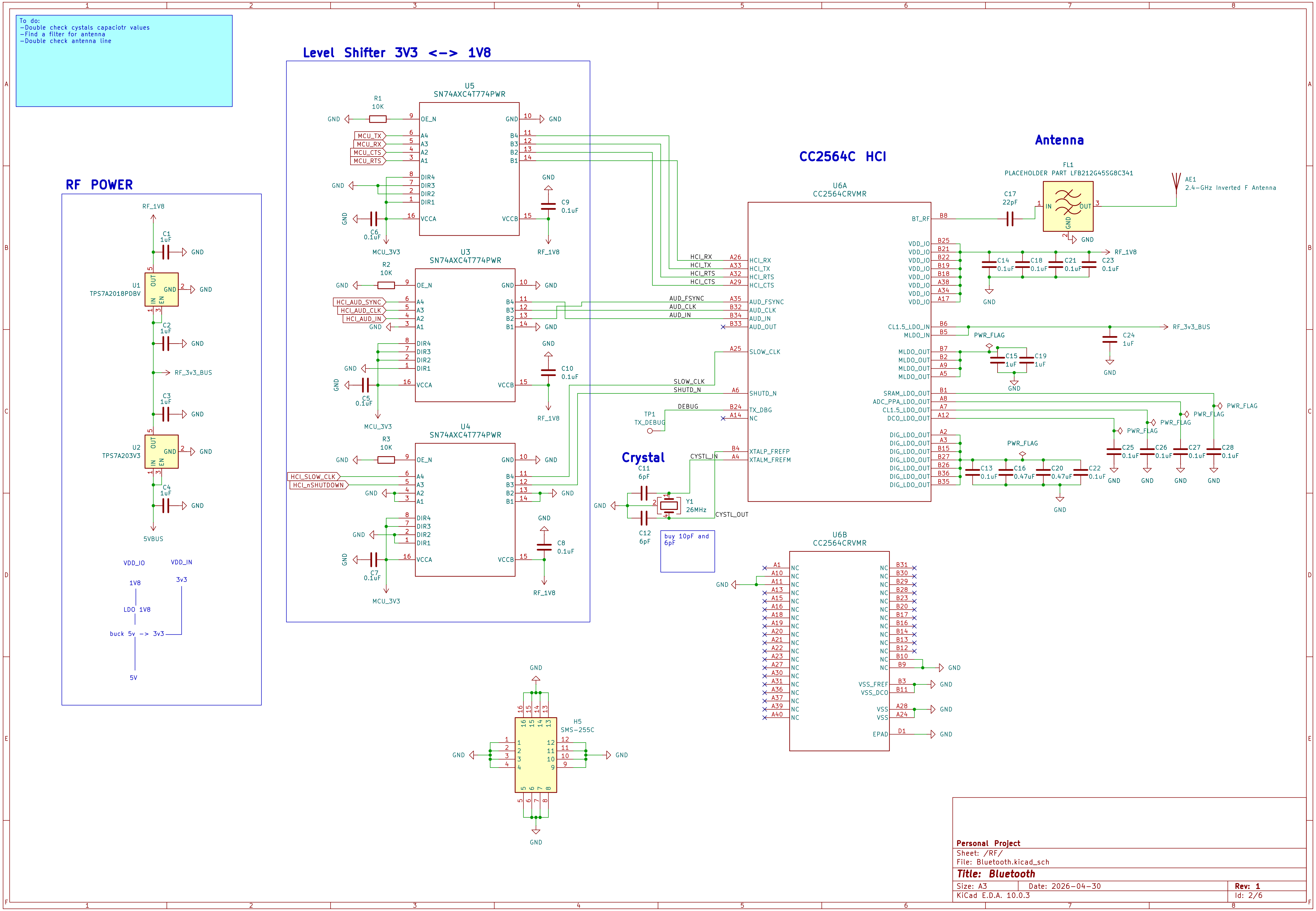
Title: Overview

Size: A4	Date: 2026-04-30	**Rev: 1**
KiCad E.D.A. 10.0.3		Id: 1/6


To do:
-Double check cystals capaciotr values
-Find a filter for antenna
-Double check antenna line
Level Shifter 3V3 <-> 1V8
U5 SN74AXC4T774PWR
R1 10K
MCU_TX
MCU_RX
MCU_CTS
MCU_RTS
C6 0.1uF
C9 0.1uF
MCU_3V3
RF_1V8
U3 SN74AXC4T774PWR
R2 10K
HCI_AUD_SYNC
HCI_AUD_CLK
HCI_AUD_IN
C5 0.1uF
C10 0.1uF
U4 SN74AXC4T774PWR
R3 10K
HCI_SLOW_CLK
HCI_nSHUTDOWN
C7 0.1uF
C8 0.1uF
RF POWER
RF_1V8
C1 1uF
U1 TPS7A2018PDBV
C2 1uF
RF_3v3_BUS
C3 1uF
U2 TPS7A203V3
C4 1uF
5VBUS
VDD_IO
VDD_IN
1V8
3v3
LDO 1V8
buck 5v -> 3v3
5V
CC2564C HCI
U6A CC2564CRVMR
HCI_RX
HCI_TX
HCI_RTS
HCI_CTS
AUD_FSYNC
AUD_CLK
AUD_IN
SLOW_CLK
SHUTD_N
DEBUG
TP1 TX_DEBUG
CYSTL_IN
CYSTL_OUT
Antenna
FL1 PLACEHOLDER PART LFB212G45SG8C341
AE1 2.4-GHz Inverted F Antenna
C17 22pF
C14 0.1uF
C18 0.1uF
C21 0.1uF
C23 0.1uF
C24 1uF
RF_3v3_BUS
PWR_FLAG
C15 0.1uF
C19 1uF
C25 0.1uF
C26 0.1uF
C27 0.1uF
C28 0.1uF
C13 0.1uF
C16 0.47uF
C20 0.47uF
C22 0.1uF
Crystal
C11 6pF
C12 6pF
Y1 26MHz
buy 10pF and 6pF
U6B CC2564CRVMR
VSS_FREF
VSS_DCO
VSS
EPAD
H5 SMS-255C
GND
Personal Project
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Size: A3
Date: 2026-04-30
Rev: 1
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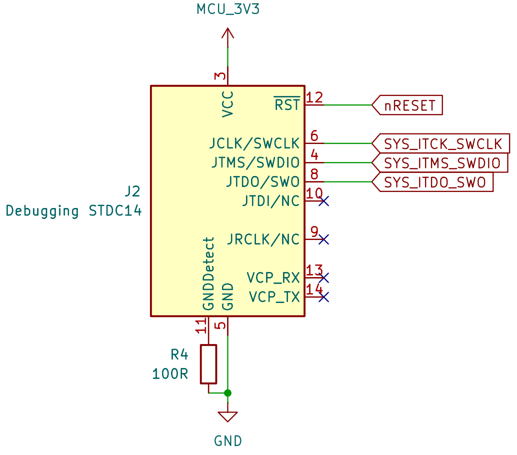
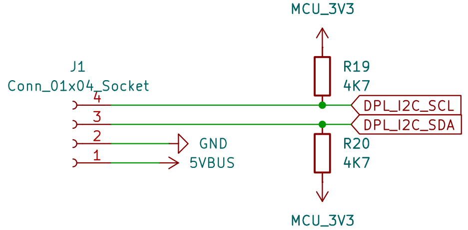
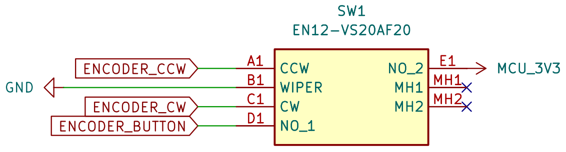
Id: 2/6



Debug Header



Display



Encoder

Personal Project

Sheet: /IO/

File: InputOutput.kicad_sch

Title: Interfacing connections and components

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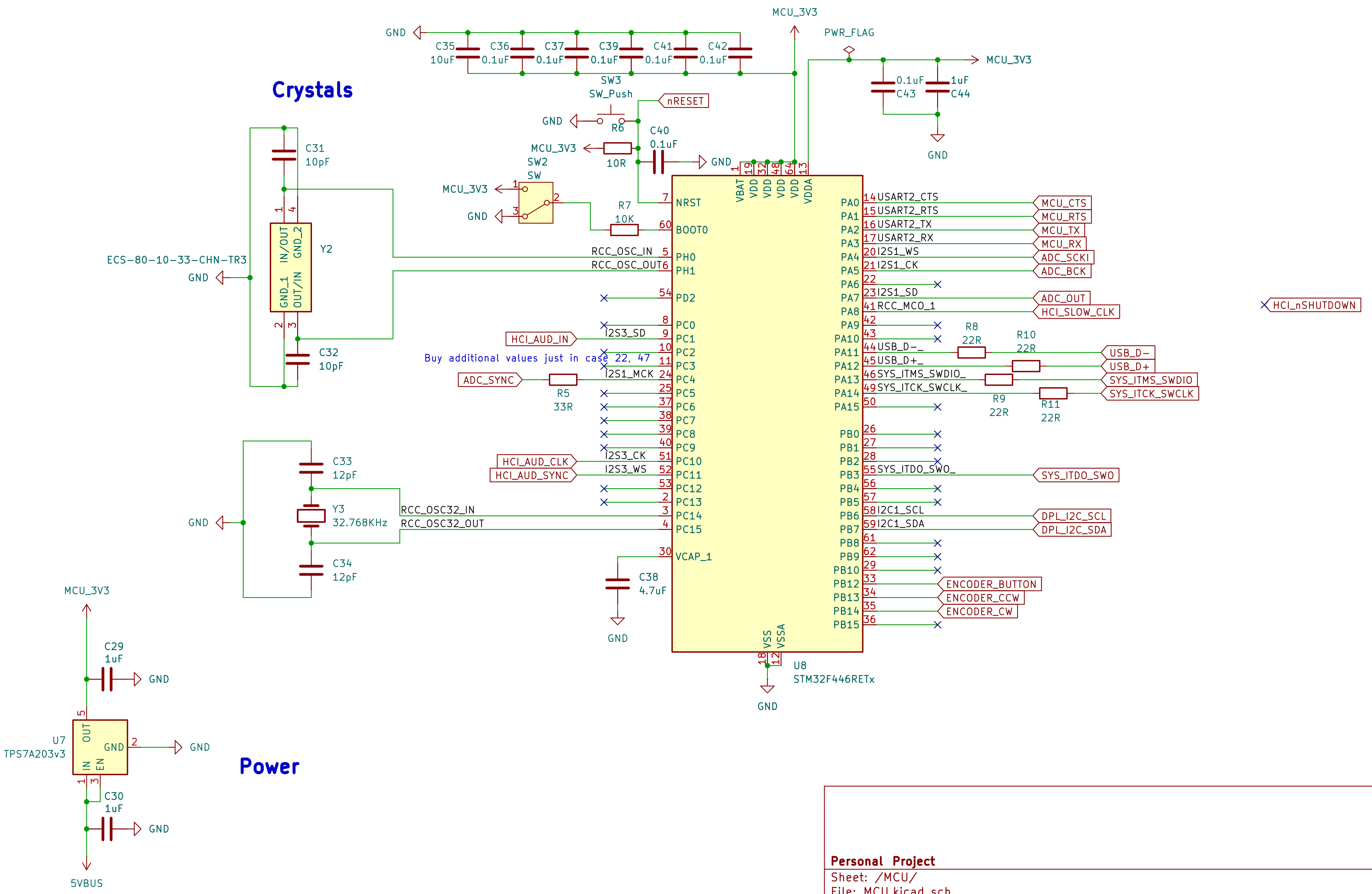

Personal Project

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Title: Microcontroller

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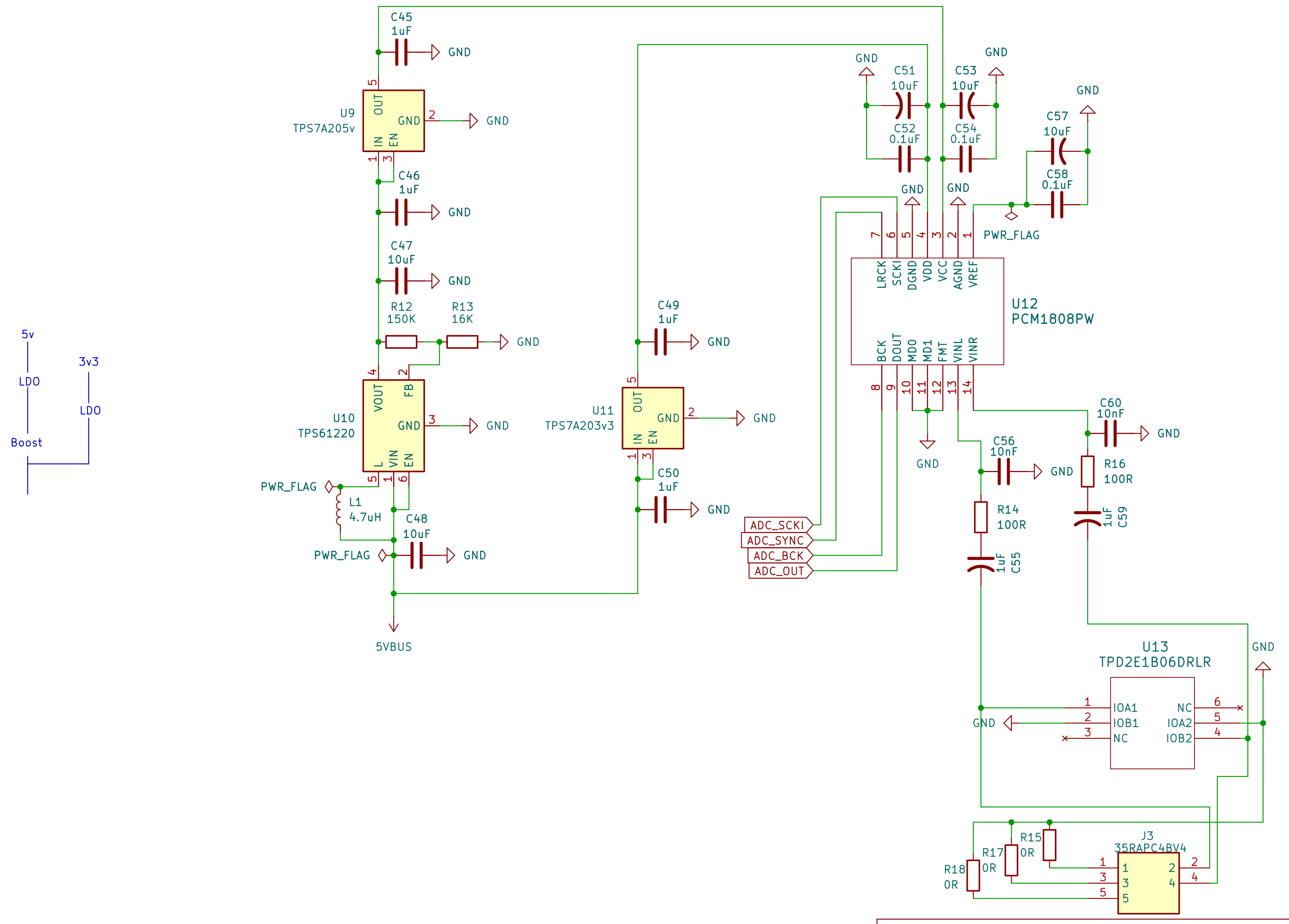

Personal Project

Sheet: /Audio/

File: Audio.kicad_sch

Title: Audio

Size: A4 | Date: 2026-04-30 | **Rev: 1**

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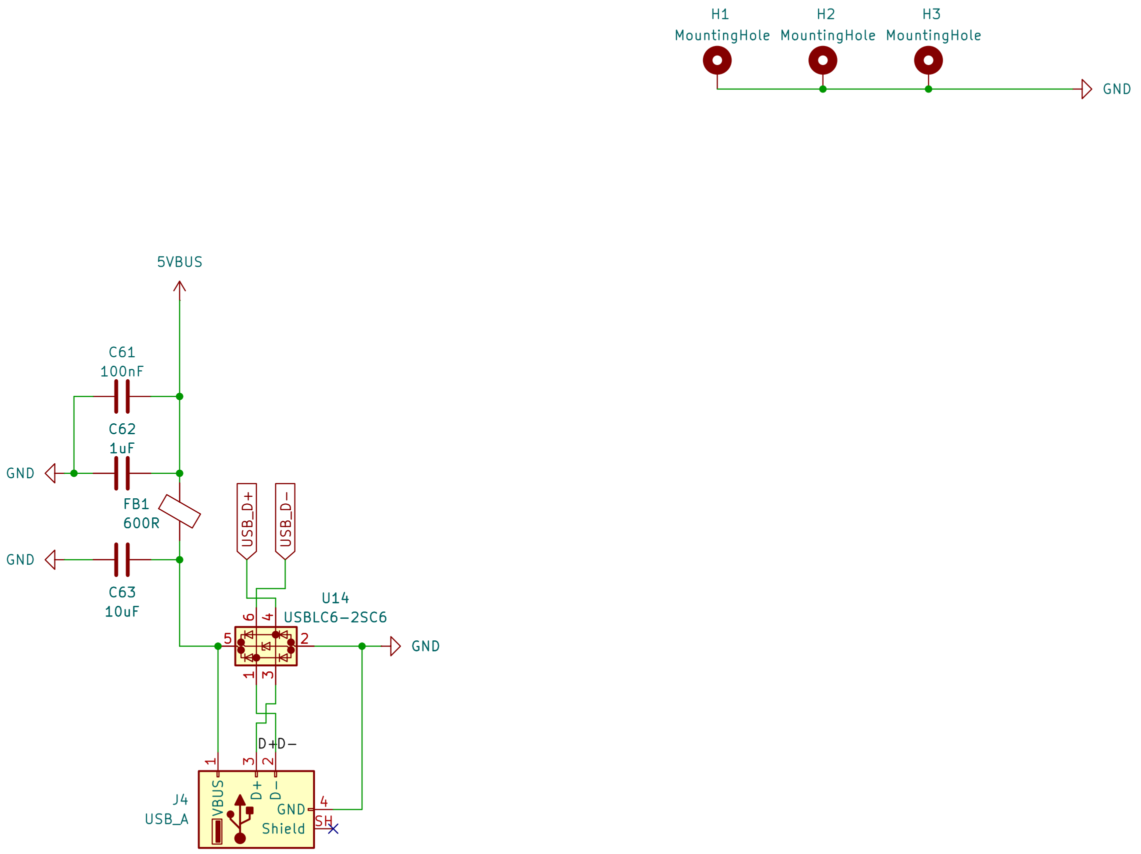

Personal Project

Sheet: /Other/

File: Other.kicad_sch

Title: USB and mounting

Size: A4	Date: 2026-04-30	Rev: 1
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